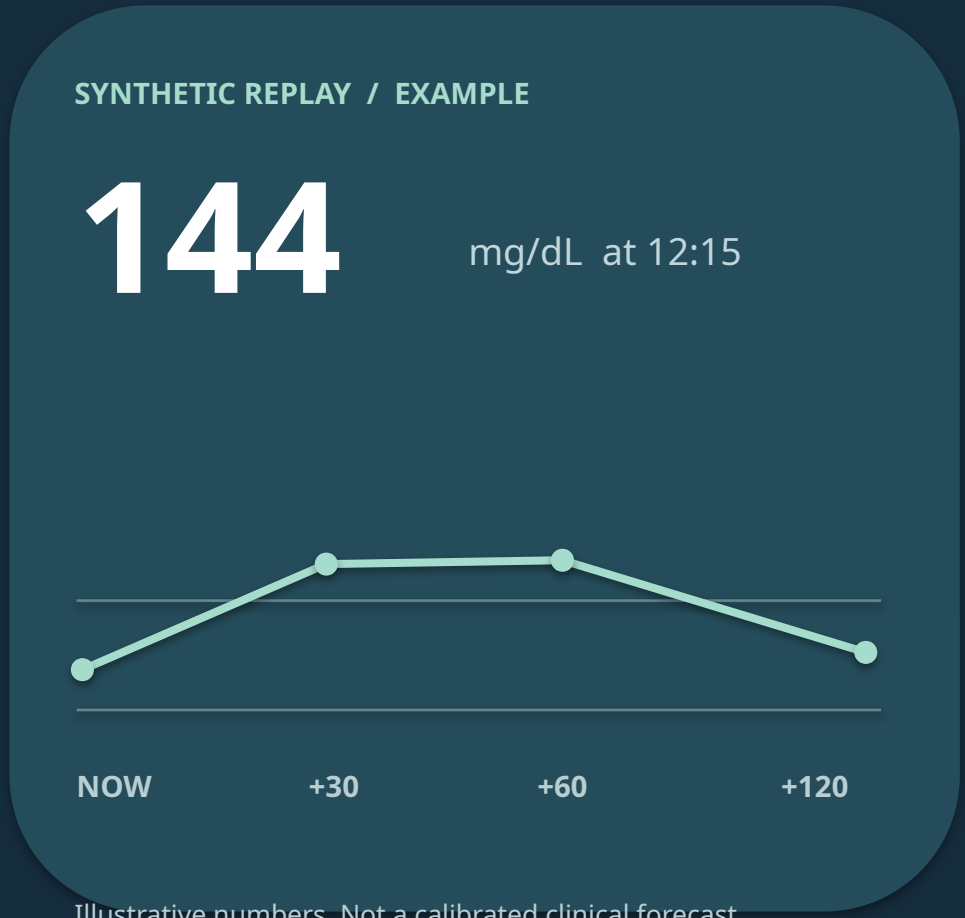
Himanshu DigitalTwin

A research prototype for near-term glucose forecasting — with a reason to withhold a prediction.

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Presenter Himanshu Shukla · MITS Gwalior



Illustrative numbers. Not a calibrated clinical forecast.

One number describes the present. It does not explain the next two hours.

Patient context and measurement quality change what a forecast can mean.

AT 12:15 PM

144

mg/dL · synthetic observed value

A reading is evidence — not a two-hour prediction.

ILLUSTRATIVE FORECAST FROM THE SAME REPLAY

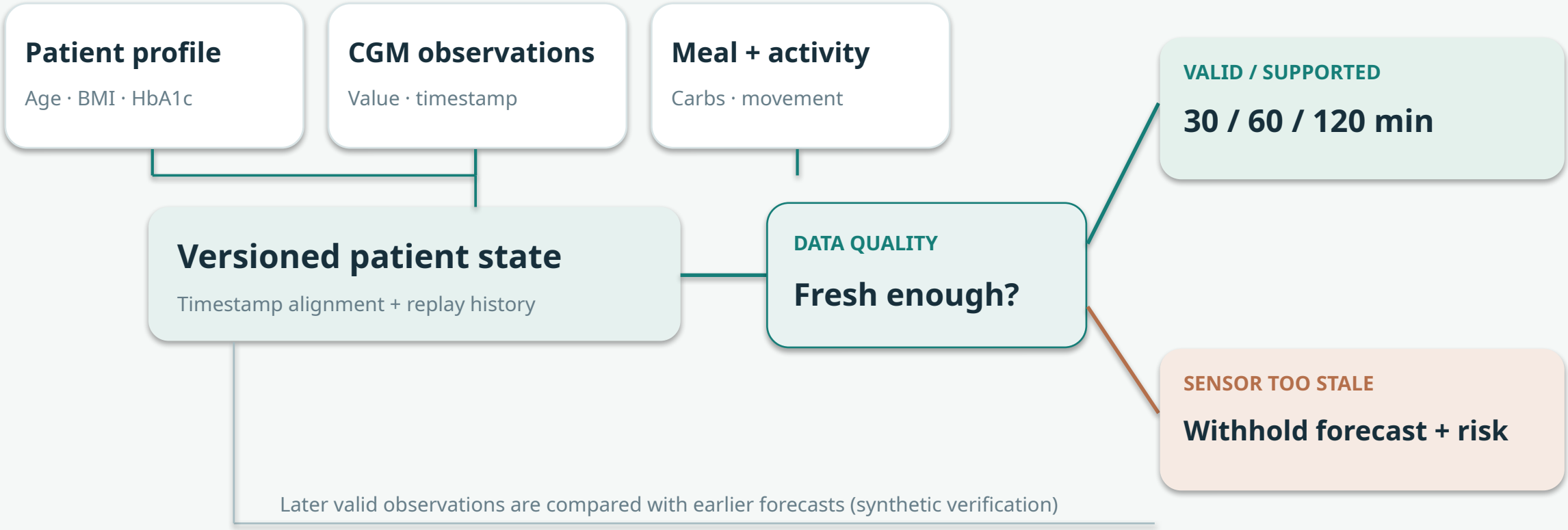
NOW	+30 min	+60 min	+120 min
144	– 155.5	– 156.0	– 145.4

Context: 59.7 g recent carbohydrates · heart rate 82 bpm · time aligned CGM

Question for the prototype: can we show useful context without hiding sensor uncertainty?

Synchronize first. Predict only when supported.

The judge can trace an observation through an explicit quality decision.



Python · FastAPI · HTML/CSS/JS / scikit-learn research pipeline is separate

Six checkpoints. One reproducible synthetic patient.

The default FastAPI dashboard lets judges jump directly to each state.

01 BASELINE
Profile + first valid CGM

02 BREAKFAST
Meal context changes

03 LUNCH
Forecast trajectory

04 VERIFICATION
Earlier forecasts resolved

05 SENSOR GAP
Abstention, no risk

06 RECOVERY
Fresh signal returns

JUDGE ACTION Run FastAPI → open localhost:8000 → Guided judge demo → choose chapter

When the signal fails, the interface must say so.

The demo includes a deliberately missing/stale glucose segment.

VALID OBSERVATION

144 mg/dL

Forecast available

Synthetic example at 12:15 PM

STALE SENSOR / 14:25

WITHHELD

No verified glucose · No forecast · No risk

Up to 15 minutes: model may use the last valid reading for a provisional estimate.

Beyond the freshness rule: no estimate is shown. Imputed values are never drawn as measured CGM.

A measured research signal. Not a deployed clinical model.

Independent CGMacros study, patient-grouped holdout; lower MAE is better.



EVIDENCE BOUNDARY

45 participants / 14 with T2D

Only 3 T2D participants in the 10-person primary holdout.

The study model is NOT the model running in the synthetic dashboard.

What the judge can verify: deterministic replay, sensor-gap abstention, and research provenance.